Rate of Reactions

Determining + Measuring Speed of Reaction

1. Volume of gas produced
- This method only works in reactions which produce a gaseous product
- Can measure ROR by observing production of gas in given amount of time

Key: Measure the volume of gas formed at regular time intervals for a fixed duration of time and plot a graph using the results
- Take readings until value becomes constant (signalling end of reaction)
- Graph must have changing gradient (not a straight line)

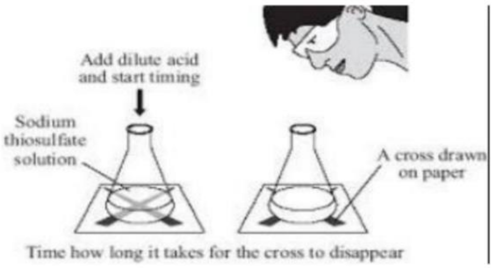
Q: explain why speed of reaction decreases with time
A: concentration/amt of *limiting* reagent is being used up
- ➔ Concentration decreases, fewer reacting particles per unit volume → particles are further apart from each other.
- ➔ Fewer collision between reacting particles per unit time
- ➔ Frequency of effective collision decreases

2. Change in Mass
- Method only works in reactions which produce gaseous products
- Can determine number of gaseous products from the loss of mas of reaction mixture → due to gas escaping from the reaction vessel

Key: measure change in mass at regular time intervals in a fixed duration of time and plot a graph using the results
- Take readings until value becomes constant (signalling end of reaction)
- Graph must have changed gradient (not straight line)

2 types: mass with time, and loss in mass with time.

3. Measuring Time taken for solid reactant to disappear OR time taken for solid product to appear
- For reactions with solid reactants / products



Calculation: Rate of Reaction = Amount of product formed / duration of time

State that you measure time for each measurement of mass/temp/volume so that you can plot a graph.

What Determines ROR

Particles are in constant random motion → will thus constantly collide with each other

Some collisions successful in breaking existing bonds and/or forming ne bonds → resulting in chemical reaction.

Definition: when a chemical reaction occurs between particles, collision is said to be an effective collision

Definition: Activation energy is the minimum amount of energy that reacting particles must possess in order for chemical reaction to proceed.

Note: for effective collision to occur → reactant particles must collide with energy that is greater than or equal to the activation energy.

- Once this condition is met → there will be effective collisions between reacting particles → resulting in product particles being formed.

Note: NEED TO SAY "FREQUENCY OF EFFECTIVE COLLISION INCREASES/DECREASES"

- Higher frequency of effective collisions between reacting particles *per unit time*→ greater amount of products formed *per unit time*→ thus faster ROR

Any factor that affects frequency of effective collisions between reacting particles will affect the speed of chemical reaction.

Concentration of Reactant(s) in Aq State

Rate of Reaction increases with an increase in concentration:

- There are more reacting particles per unit volume, the particles are closer together
- Resulting in *more collision* between reacting particles per unit time
- Thus, frequency of effective collision increases
- ROR increases

Rate of Reaction decreases with decrease in concentration

- There are fewer reacting particles per unit volume, particles are further apart
- Resulting in *fewer collisions* between reacting particles per unit time.

Pressure at which reaction occurs

- only for reactions involving at least 1 gaseous product
- for gasses we can't use the term concentration, thus we use the term pressure
- pressure → very little effect on reactants in solid, liquid, Aq state → as particles in these substances are alr very closely packed together.

Rate of Reaction increases with an increase in pressure:

- reacting particles are now closer together
- thus, there are more reacting particles per unit volume
- resulting in more collisions between reacting particles per unit time
- thus, frequency of effective collision increases.

- Thus, frequency of effective collision decreases
- ROR decreases

Ensure that you state what the reacting particles are for clarity. If you have the volume state that too.

- ROR increases.

Rate of Reaction decreases with decrease in pressure

- Reacting particles are further apart
- Thus, there are fewer reacting particles per unit volume
- Resulting in fewer collisions between reacting particles per unit time
- Thus, frequency of effective collision decreases
- ROR decreases

<u>Particle size of reactant</u>

Rate of Reaction increases with an increase in particle size:

- There is larger total surface area exposed of the reacting particles to collide into
- Thus, resulting in more collisions between reacting particles per unit time
- Thus, frequency of effective collision increases.
- ROR increases

Rate of Reaction decreases with decrease in particle size:

- There is lesser total exposed surface area of reacting particles to collide into
- Thus, resulting in fewer collisions between reacting particles per unit time
- Thus, frequency of effective collision decreases
- ROR decreases

➔ do not write SA to V ratio

➔ data logger can be used when reaction is too vigorous for data to be collected manually

<u>Temperature at which reaction occurs</u>

- Usually, temp. of reactants that are Aq solutions
- 2 key points – 1 about kinetic energy increase causing increase in collision, and other about the fact that more reacting particles have energy equal to or greater than activation energy

Rate of Reaction increases with an increase in temperature

- Reacting particles gain more kinetic energy, move/vibrate faster and collide with each other more often per unit time and collide with more energy
- More reacting particles have energy greater than or equal to activation energy
- Thus, frequency of effective collision increases
- ROR increases.

Rate of reaction decreases with a decrease in temperature

- Reacting particles lose kinetic energy, move/vibrate slower and collide with each other less often per unit time, and collide with less energy
- Fewer reacting particles have energy greater than or equal to the activation energy

- ➔ particle size of catalysed used could be compared instead of particle size of reactant.
- ➔ Just write powdered form in exam.

- Frequency of effective collision decreases.
- ROR decreases.

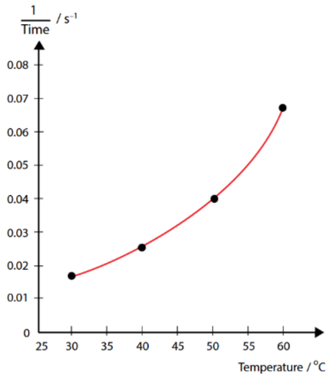
Example:

Sodium Thiosulfate + HCl → Sodium Chloride + water + Sulfur dioxide + Sulfur

- Yellow precipitate of sulfur is produced. This reduces amount of light passing through solution
- Hence, we can find ROR by measuring time taken for solution to turn opaque

Rate of reaction is inversely proportional to the time taken.

By plotting a graph of 1/time with temperature → we can directly observe how rate of reaction varies with temperature.



Presence of Catalyst

2 important points for catalysts:

1. Catalysts provide an alternative pathway of lower activation energy for the reaction to occur. This speed up ROR
2. More reacting particles now have energy greater than or equal to activation energy. Thus, the frequency of effective collisions between reacting particles increases, hence the rate of reaction increases.

Catalysts DO NOT cause reacting particles to collide more often

Other possible factors:

- Basicity of acids (link to concentration)
- Strength of acids (link to concentration)
- Reactivity of metals
- Thermal stability of metal carbonates
- More initial mass of reagent added
- layer of insoluble salt formed
- layer of aluminium oxide

- Catalysts provide alternative pathway of lower activation energy → for both forward reaction and reverse reaction of a reversible reaction - Catalysts speed up chemical reactions without themselves being used up in reactions → they are very useful and efficient. Example: Explain why presence of finely divided iron increases ROR between nitrogen and hydrogen gas to produce ammonia gas • finely divided iron is a catalyst which provides an alternative pathway with a lower activation energy for the reaction to proceed • more reacting particles (of nitrogen and hydrogen) now have energy equal to or greater than the activation energy. • Thus, frequency of effective collisions between reacting particles (of nitrogen and hydrogen gas) increases and the rate of reaction also increases	Example: Magnesium metal reacts faster with 1.0mol/dm3 of sulfuric acid compared to 1.0mol/dm3 of hydrochloric acid • HCl → monobasic acid → 1H+ ion for every molecule upon ionization in water. • Sulfuric acid → dibasic acid → 2H+ ions when each molecule of acid ionizes in water. • higher concentration of H+ ions in 1mol/dm3 Sulfuric acid compared to 1mol/dm3 Hydrochloric acid, by 2 times. Thus, Sulfuric acid is a stronger acid than HCl. • - Number of H+ ions per unit volume is higher → thus there are more collisions between Mg and H+ ions, which causes increase in frequency of effective collision between these particles. • Thus, faster reaction with sulfuric acid compared to hydrochloric acid

All about Catalysts!

Definition: Catalyst is a substance that increase ROR by providing an alternative pathway with a lower activation energy for the reaction to proceed. It remains chemically unchanged at the end of the reaction

<u>Characteristics of Catalysts:</u>

- Provides alternative pathway which has lower activation energy for the reaction to proceed
  - Original pathway with higher activation energy is still available
- Catalysts only change speed (increase ROR) but does not affect yield of chemical reaction
  - Final amt of products formed after reaction remains same
  - Only affects time taken to form final amt of product
- Only small amount of catalyst needed to change ROR

- Remains chemically unchanged at the end of the reaction, but physical appearance might change.
    - Samt amount of catalyst present at the beginning and at the end of reaction
    - Possible for catalyst to undergo chemical changes at an intermediate state of reaction → but will be reformed at end of reaction
- They are selective → only catalyse certain reactions
- Impurities can poison/inactivate them → then they cannot catalyse reactions

Note: Catalysts are often transition metals or their compounds

Industrial Catalysts

- Industrial manufacture of ammonia by Haber Process → catalysed by finely divided iron → allowing good yields at low pressure

- Petroleum refining → large hydrocarbon molecules in crude oil → need to be broken down into smaller molecules.
- This is so that they can be used as fuels + also as raw materials to produce plastics
- The breaking of chemical bonds proceeds very slowly even with high heat and pressure
- By using catalysts → less extreme physical conditions required. This saves energy and makes refinery safer to operate

- Similarly→ production of plastics benefit from catalysts
- This is because formation of long chain of polymers → would proceed very slowly without use of catalysts

Example:
ethane($H2H4$) and hydrogen react to give ethane($C2H6$)
This reaction occurs extremely slowly even at high temp and pressure → but when nickel is added as a catalyst → reaction occurs readily at 200 degrees
Speed of above reaction increases in presence of nickel → as nickel provides alternative pathway with lower activation energy for the reaction to proceed.

Sample:
Explain why presence of finely divided iron increases ROR between nitrogen and hydrogen gas to produce ammonia gas

- Finely divided iron is a catalyst which provides alternative pathway with a lower activation energy for the reaction to proceed.
- More reacting particles (of nitrogen and hydrogen gas) now have energy equal to or greater than the activation energy
- Thus, frequency of effective collisions between reacting particles (of hydrogen and nitrogen gas) increases and the ROR also increases.

some laundry detergents have enzymes added to break down oil/other stains → that would otherwise require high heat or strong chemicals to remove.
enzymes operate at optimally milder temp → stubborn stains can be removed using these detergents with warm water.

There are also biological catalysts → which I am skipping as it is part of biology so you should already know it.

Note: volume of products formed is different only if:

- Acid is the limiting reactant, and the other reactant is in excess

- If concentration is more, regardless of mol, the ROR is greater.